

QuantaPlex Series T42S-2U/T42SP-2U

**2U 4-Node Server Featuring
Latest Cloud Computing Technology**

User's Guide

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TABLE OF CONTENT

About the System

Introduction	1-1
System Features.....	1-1
.....	1-3
Package Contents	1-4
A Tour of the System	1-5
System Overview	1-5
2.5" HDD System.....	1-5
3.5" HDD System.....	1-6
System Front View.....	1-7
2.5" HDD SKU	1-7
3.5" HDD SKU	1-8
Front Control Panel and Status LED	1-9
Rear View	1-10
Rear I/O Ports	1-10
HDD Configuration	1-11
2.5" HDD Configuration	1-11
3.5" HDD Configuration	1-12
Power Sub-System	1-12
PSU LED	1-13
Cooling Sub-System.....	1-14
Enabling the Intel® VMD technology	1-15
Enabling Platform Trusted Technology (PTT)	1-16
TPM configuration settings	1-17

Regulatory and Compliance Information

Conventions

Several different typographic conventions are used throughout this manual. Refer to the following examples for common usage.

Bold type face denotes menu items, buttons and application names.

Italic type face denotes references to other sections, and the names of the folders, menus, programs, and files.

<Enter> type face denotes keyboard keys.



WARNING!

Warning information appears before the text it references and should not be ignored as the content may prevent damage to the device.



CAUTION!

CAUTIONS APPEAR BEFORE THE TEXT IT REFERENCES, SIMILAR TO NOTES AND WARNINGS. CAUTIONS, HOWEVER, APPEAR IN CAPITAL LETTERS AND CONTAIN VITAL HEALTH AND SAFETY INFORMATION.

Note:

Highlights general or useful information and tips.

Precautionary Measures

Read all caution and safety statements in this document before performing any of the instructions. To reduce the risk of bodily injury, electrical shock, fire, and equipment damage, read and observe all warnings and precautions in this chapter before installing or maintaining your system. To avoid personal injury or property damage, before you begin installing the product, read, observe, and adhere to all of the following instructions and information. The following symbols may be used throughout this guide and may be marked on the product and / or the product packaging.

Safety Instructions about your system

In the event of a conflict between the information in this guide and information provided with the product or on the website for a particular product, the product documentation takes precedence.

Your system should be integrated and serviced only by technically qualified persons.

You must adhere to the guidelines in this guide and the assembly instructions in related chapters to ensure and maintain compliance with existing product certifications and approvals. Use only the described, regulated components specified in this guide. Use of other products / components will void the UL Listing and other regulatory approvals of the product, and may result in noncompliance with product regulations in the region(s) in which the product is sold.

Table 1: Warning and Cautions

CAUTION	Indicates the presence of a hazard that may cause minor personal injury or property damage if the CAUTION is ignored.
WARNING	Indicates the presence of a hazard that may result in serious personal injury if the WARNING is ignored.
	Indicates potential hazard if indicated information is ignored.
	Indicates shock hazards that result in serious injury or death if safety instructions are not followed.
	Indicates hot components or surfaces.
	Indicates do not touch fan blades, may result in injury.
	Remove the system from the rack to disconnect power system.

Table 1: Warning and Cautions (Continued)

	The enclosure is designed to carry only the weight of the system sled. Do not use this equipment as a workspace. Do not place additional load onto any equipment in this system.
	Indicates two people are required to safely handle the system.
	Restricted Access Location: The system is intended for installation only in a Server Room or Computer Room where both these conditions apply: • access can only be gained by SERVICE PERSONS or by USERS who have been instructed about the reasons for the restrictions applied to the location and about any precautions that shall be taken; and • access is through the use of a TOOL or lock and key, or other means of security, and is controlled by the authority responsible for the location.

Intended Application Uses

This product was evaluated as Information Technology Equipment (ITE), which may be installed in offices, schools, computer rooms, and similar commercial type locations. The suitability of this product for other product categories and environments (such as medical, industrial, residential, alarm systems, and test equipment), other than an ITE application, may require further evaluation.

Site Selection

The system is designed to operate in a typical office environment. Choose a site that is:

- Clean, dry, and free of airborne particles (other than normal room dust).
- Well-ventilated and away from sources of heat including direct sunlight and radiators.
- Away from sources of vibration or physical shock.
- Isolated from strong electromagnetic fields produced by electrical devices.
- In regions that are susceptible to electrical storms, we recommend you plug your system into a surge suppressor and disconnect telecommunication lines to your modem during an electrical storm.
- Provided with a properly grounded wall outlet.
- Provided with sufficient space to access the power system, because they serve as the product's main power disconnect.
- Provided with either two independent DC power system or two independent phases from a single power system.

Equipment Handling Practices

Reduce the risk of personal injury or equipment damage:

- Conform to local occupational health and safety requirements when moving and lifting equipment.
- Use mechanical assistance or other suitable assistance when moving and lifting equipment.
- To reduce the weight for easier handling, remove any easily detachable components.
- Never lift or move your system solely by the handle on the component.

Power and Electrical Warnings



CAUTION!

MAKE SURE THE SYSTEM IS REMOVED FROM THE RACK BEFORE SERVICING ANY NON-HOT PLUG COMPONENTS. THE BUS BAR CLIPS MUST BE DISCONNECTED FROM THE POWER SYSTEM IN ORDER TO FULLY SEPARATE THE SYSTEM FROM THE POWER SOURCE.



CAUTION!

TO AVOID RISK OF ELECTRIC SHOCK, DISCONNECT ALL CABLING FROM THE SYSTEM AND REMOVE THE SYSTEM FROM THE RACK.

System Access Warnings



CAUTION!

TO AVOID PERSONAL INJURY OR PROPERTY DAMAGE, THE FOLLOWING SAFETY INSTRUCTIONS APPLY WHENEVER ACCESSING THE INSIDE OF THE PRODUCT:

- Disconnect from the power source by removing the system from the rack.
- Disconnect all cabling running into the system.
- Retain all screws or other fasteners when servicing. Upon completion servicing, secure with original screws or fasteners.



CAUTION!

IF THE SERVER HAS BEEN RUNNING, ANY INSTALLED HDD MODULES MAY BE HOT.



CAUTION!

UNLESS YOU ARE ADDING OR REMOVING A HOT-PLUG COMPONENT, ALLOW THE SYSTEM TO COOL BEFORE SERVICING.



CAUTION!

TO AVOID INJURY DO NOT CONTACT MOVING FAN BLADES. IF YOUR SYSTEM IS SUPPLIED WITH A GUARD OVER THE FAN, DO NOT OPERATE THE SYSTEM WITHOUT THE FAN GUARD IN PLACE.

Rack Mount Warnings

The following installation guidelines are required by UL for maintaining safety compliance when installing your system into a rack.

The equipment rack must be anchored to an unmovable support to prevent it from tipping when your system or piece of equipment is extended from it. The equipment rack must be installed according to the rack manufacturer's instructions.

Install equipment in the rack from the bottom up, with the heaviest equipment at the bottom of the rack.

Extend only one piece of equipment from the rack at a time.

You are responsible for installing a main power disconnect for the entire rack unit. This main disconnect must be readily accessible, and it must be labeled as controlling power to the entire unit, not just to the system(s).

To avoid risk of potential electric shock, a proper safety ground must be implemented for the rack and each piece of equipment installed in it.

Elevated Operating Ambient - If installed in a closed or multi-unit rack assembly, the operating ambient temperature of the rack environment may be greater than room ambient. Therefore, consideration should be given to installing the equipment in an environment compatible with the maximum ambient temperature (T_{ma}) specified by the manufacturer.

Reduced Air Flow - Installation of the equipment in a rack should be such that the amount of air flow required for safe operation of the equipment is not compromised.

Mechanical Loading - Mounting of the equipment in the rack should be such that a hazardous condition is not achieved due to uneven mechanical loading.

Circuit Overloading - Consideration should be given to the connection of the equipment to the supply circuit and the effect that overloading of the circuits might have on over-current protection and supply wiring. Appropriate consideration of equipment nameplate ratings should be used when addressing this concern.

Reliable Earthing - Reliable earthing of rack-mounted equipment should be maintained.

Particular attention should be given to supply connections other than direct connections to the branch circuit (e.g. use of power strips).

Electrostatic Discharge (ESD)



CAUTION!

ESD CAN DAMAGE DRIVES, BOARDS, AND OTHER PARTS. WE RECOMMEND THAT YOU PERFORM ALL PROCEDURES AT AN ESD WORKSTATION. IF ONE IS NOT AVAILABLE, PROVIDE SOME ESD PROTECTION BY WEARING AN ANTI-STATIC WRIST STRAP ATTACHED TO CHASSIS GROUND -- ANY UNPAINTED METAL SURFACE -- ON YOUR SERVER WHEN HANDLING PARTS.

Always handle boards carefully. They can be extremely sensitive to ESD. Hold boards only by their edges without any component and pin touching. After removing a board from its protective wrapper or from the system, place the board component side up on a grounded, static free surface. Use a conductive foam pad if available but not the board wrapper. Do not slide board over any surface.

Cooling and Airflow



CAUTION!

CAREFULLY ROUTE CABLES AS DIRECTED TO MINIMIZE AIRFLOW BLOCKAGE AND COOLING PROBLEMS. FOR PROPER COOLING AND AIRFLOW, OPERATE THE SYSTEM ONLY WITH THE CHASSIS COVERS* / AIR DUCT INSTALLED. OPERATING THE SYSTEM WITHOUT THE COVERS / AIR DUCT IN PLACE CAN DAMAGE SYSTEM PARTS . TO INSTALL THE COVERS* / AIR DUCT:

- Check first to make sure you have not left loose tools or parts inside the system.
- Check that cables, add-in cards, and other components are properly installed.

Attach the covers* / air duct to the chassis according to the product instructions.

* May not apply to all systems.

Please be aware that slots and openings on the front and rear side of the chassis are designed for ventilation; to make sure reliable operation of your system and to protect it from overheating, these openings must not be covered or blocked. The openings should never be covered or blocked by placing the product on a bed, sofa, rug, or other similar surface. This product should never be placed near or over a radiator or heat register, or in a built-in installation unless proper ventilation is provided.

Laser Peripherals or Devices



CAUTION!

TO AVOID RISK OF RADIATION EXPOSURE AND / OR PERSONAL INJURY:

- Do not open the enclosure of any laser peripheral or device.
- Laser peripherals or devices are not serviceable.
- Return to manufacturer for servicing.

Use certified and rated Laser Class I for Optical Transceiver product.

Heed safety instructions: Before working with the system, whether using this manual or any other resource as a reference, pay close attention to the safety instructions. Adhere to the assembly instructions in this manual to ensure and maintain compliance with existing product certifications and approvals. Use only the described, regulated components spec-

ified in this manual. Use of other products / components will void the UL listing and other regulatory approvals of the product and will most likely result in non-compliance with product regulations in the region(s) in which the product is sold.

System power on/off: To remove power from system, you must remove the system from rack. Make sure the system is removed from the rack before opening the chassis, adding, or removing any non hot-plug components.

Hazardous conditions, devices and cables: Hazardous electrical conditions may be present on power, telephone, and communication cables. Turn off the system and disconnect the cables attached to the system before opening it. Otherwise, personal injury or equipment damage can result.

Electrostatic discharge (ESD) and ESD protection: ESD can damage drives, boards, and other parts. We recommend that you perform all procedures in this chapter only at an ESD workstation. If one is not available, provide some ESD protection by wearing an antistatic wrist strap attached to chassis ground any unpainted metal surface on the server when handling parts.

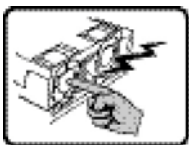
ESD and handling boards: Always handle boards carefully. They can be extremely sensitive to electrostatic discharge (ESD). Hold boards only by their edges. After removing a board from its protective wrapper or from the server, place the board component side up on a grounded, static free surface. Use a conductive foam pad if available but not the board wrapper. Do not slide board over any surface.

Installing or removing jumpers: A jumper is a small plastic encased conductor that slips over two jumper pins. Some jumpers have a small tab on top that can be gripped with fingertips or with a pair of fine needle nosed pliers. If the jumpers do not have such a tab, take care when using needle nosed pliers to remove or install a jumper; grip the narrow sides of the jumper with the pliers, never the wide sides. Gripping the wide sides can damage the contacts inside the jumper, causing intermittent problems with the function controlled by that jumper. Take care to grip with, but not squeeze, the pliers or other tool used to remove a jumper, or the pins on the board may bend or break.

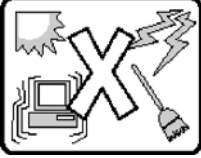
General Information

Before servicing this system, it is recommended to read this guide completely to be aware of any safety issues or requirements involved in the servicing of this system.

Assembly Safety Guidelines



The power system in this product contains no user-serviceable parts.
Refer servicing only to qualified personnel.

	The system is designed to operate in a typical office environment. Choose a site that is: • Clean and free of airborne particles (other than normal room dust). • Well ventilated and away from sources of heat including direct sunlight. • Away from sources of vibration or physical shock. • Isolated from strong electromagnetic fields produced by electrical devices. • In regions that are susceptible to electrical storms, we recommend you plug your system into a surge suppressor and disconnect telecommunication lines to your modem during an electrical storm. • Provided with a properly grounded wall outlet. • Provided with sufficient space to access the power system, because they serve as the product's main power disconnect.
	WARNING! The system is safety certified as rack-mounted equipment for use in a server room or computer room, using an approved customer rack. The enclosure is designed to carry only the weight of the system sled. Do not place additional load onto any equipment.
	Heavy object. Indicates two people are required to safely handle the system.

Structure of this guide

- Chapter 1: About the System
“This section introduces the system, its different configuration(s) and the main features.”
- Chapter 2: Regulatory and Compliance Information
“This section provides regulatory and compliance information applicable to this system.”

About the System

Chapter 1

This section introduces the system, its different configuration(s) and the main features.

1.1 Introduction

This document provides an overview of the hardware features of the chassis, troubleshooting information, and instructions on how to add and replace components of the server.

For the latest version of this manual, see www.QCT.io.

System Features

The system comprises a 2U/31.1" long chassis. Major features include:

- **T42S-2U:** available for both 3.5" SKU and 2.5" SKU
- **T42SP-2U:** only available for 2.5" SKU
- **Chipset:** Intel® C621 / C624 series
- **Processors (x2):** Intel® Xeon® Processor Scalable Family (codename Skylake-SP)
- **PCIe slots:**
 - (1) PCIe x16 Gen3 riser slot for PCIe MD2 card (CPU1)
 - (1) PCIe x8 Gen3 mezzanine slot for LSI 3008A/3108A card (CPU0)
 - (1) PCIe x16 Gen3 OCP mezzanine slot for OCP card (CPU0)
- **Memory:** Up to 16 DIMM slots are available; ECC DDR4 2666 MT/s RDIMM memory
- **Network*:**
 - Dedicated GbE management NIC port from PHY RTL8211 to BMC
 - Intel® C621 as 4x GbE integrated network solution with PHY (optional)
 - Intel® C624 as 4x 10GbE integrated network solution with PHY (optional)

*Visit www.QCT.io for the latest Network support listings.

Note:

The system supports:

1600W PSU: 200-240Vac, 50/60Hz, 10A or 240Vdc, 10A

2200W PSU: 100-120Vac, 50/60Hz, 15.5A Max or 240Vdc, 14A Max

Specifications

Table 1.1: System Specifications

SPECIFICATIONS	DESCRIPTION
Form factor	2U rack mount

Table 1.1: System Specifications (Continued)

SPECIFICATIONS	DESCRIPTION
Chassis dimensions (W x H x D)	444mm x 87.5 mm x 790 mm 17.48" x 3.44" x 31.1"
Mainboard form factor (W x L), half-width	165mm x 508mm / 6.5" x 20" - Up to 4 independent mainboard sleds 2.5" SKU - 6 x 2.5" HDD per node 3.5" SKU - 3 x 3.5" HDD per node
Processor	Processor type: Intel® Xeon® Processor Scalable Family (codename Skylake-SP) Max. TDP support: 165W, Optimized power delivery for 85W, VRD 13 Number of processors: 2 Internal Interconnect: 10.4 GT/s, 9.6 GT/s
Chipset	Intel® C621 / C624
Memory	Total slots: 16 Memory type: DDR4 2666 MT/s RDIMM Memory size: 8GB, 16GB, 32 GB* *More options refer to the AVL
Storage controller	Onboard (Intel® C621 / C624): (4) SATA 6Gbps port with (1) slimline connector (4) sSATA 6Gbps port with (1) slimline connector
Networking	- Dedicated GbE management NIC port from PHY RTL8211 to BMC - Intel® C621 as 4x GbE Integrated Network Solution with PHY (optional) - Intel® C624 as 4x10GbE Integrated Network Solution with PHY (optional)
Expansion slots	(1) PCIe x16 Gen3 riser slot for PCIe MD2 card (CPU0) (1) PCIe x8 Gen3 mezzanine slot for LSI 3008A/3108A card (CPU0) (1) PCIe x16 Gen3 OCP mezzanine slot for OCP card (CPU0)
Storage	(12) 3.5" hot-plug HDD for system, (3) 3.5" hot-plug per node or (24) 2.5" hot-plug HDD for system, (6) 2.5" hot-plug with (2) NVMe* per node (1) M.2 2280 PCIe x2 / SATA per node * Only for model T42SP-2U
Onboard storage	(2) SATADOM (optional)
HDD backplane	1 to 1
Video	Integrated Aspeed AST2500 with 8MB DDR4 video memory
HW RAID options	- LSI® 3008A 12Gb/s RAID mezzanine, RAID 0, 1, 5, 10. RAID 6 with additional RAID key - LSI® 3108A 12Gb/s RAID mezzanine, RAID 0, 1, 5, 10. RAID 6 with additional RAID key
Network options	(1) GbE quad port OCP mezzanine PHY card (Optional)* (1) 10GbE quad port OCP mezzanine PHY card (Optional)* - More options refer to the AVL at www.qct.io *The quad port PHY card is Installed to OCP mezzanine slot

Table 1.1: System Specifications (Continued)

SPECIFICATIONS	DESCRIPTION
Front I/O	(1) USB 2.0 port per node
Rear I/O	• (2) USB 3.0 ports per node • (1) VGA per node • (1) Micro-USB port per node • (1) MicroSD slot per node • (1) GbE RJ45 management port per node
TPM	Yes (optional, SPI mode)
ACPI	ACPI compliance, S0, S5 support
Power supply	• (2) 1600W high efficiency hot-plug redundant PSUs, 200-240Vac, 50/60Hz, 10A or 240Vdc, 10A • (2) 2200W high efficiency hot-plug redundant PSUs, 100-120Vac, 50/60Hz, 15.5A Max or 240Vdc, 14A Max
System rating	• 100-120/200-240Vac, 50/60Hz, 10A or 240Vdc, 8A (Per PSU inlet) • 200-240Vac, 50/60Hz, 10A or 240Vdc, 8A (Per PSU inlet)
Fan	(4) hot-plug* dual-rotor fan modules *Hot-plug feature is supported only when Cable Management Arm installed
System management	IPMI v2.0 Compliant, on board "KVM over IP" support
Operating environment	• Operating temperature: 5°C to 35°C (41°F to 95°F) • Non-operating temperature: -40°C to 70°C (-40°F to 158°F) • Operating relative humidity: 20% to 85%RH • Non-operating relative humidity: 10% to 95%RH

1.2 Package Contents

- (1) T42S-2U/T42SP-2U system
- (2) processor heat sinks per node
- (2) power supply unit
- (2) power cord (optional)

Note:

Note: For exact shipping contents, contact your sales representative.

1.3 A Tour of the System

System Overview

The system is available in two models: 24 x 2.5" HDD or 12 x 3.5" HDD model. The following illustrations describe the major components of these two variants.

2.5" HDD System

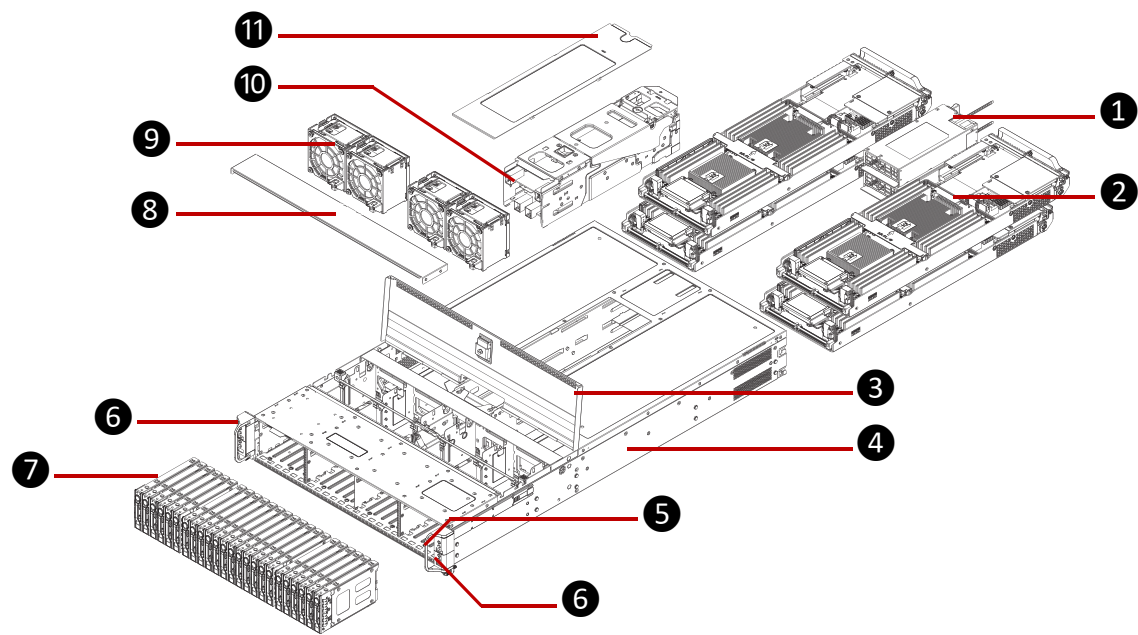


Figure 1-1. 2.5" HDD System Component Overview

Table 2: 2.5" System Component Overview

No.	ITEM	DESCRIPTION
1	Power supply unit (PSU)	Two 1600W high efficiency redundant PSU, 200-240Vac, 50/60Hz, 10A or 240Vdc, 10A
		Two 2200W high efficiency redundant PSU, 100-120Vac, 50/60Hz, 15.5A Max or 240Vdc, 14A Max
2	MB sled	Hot-swap mainboard sleds (x4)
3	Fan assembly cover	Open to service fan modules
4	Chassis	2U System chassis 444mm/17.48"(W)x 87.5mm/3.44"(H)x 790mm/31.1"(D)
5	Server handle	Two server handles used for pulling the system out of the rack CAUTION! THE HANDLES ARE DESIGNED FOR THE EXTENSION OF THE SYSTEM FROM THE RACK. THE HANDLES ARE NOT DESIGNED TO CARRY THE WEIGHT OF THE SYSTEM. DO NOT USE THE HANDLES TO MOVE OR LIFT THE SYSTEM.

Table 2: 2.5" System Component Overview (Continued)

No.	ITEM	DESCRIPTION
6	Front control panels	Front control panels 1/2 (left) and 3/4 (right). See <i>Front Control Panel and Status LED</i> on page 1-9 for further information
7	2.5" HDD bay	2.5" hard disk drive (HDD) modules (x24)
8	Neck-top cover	Remove to service HDD backplane. Only available in 2.5" HDD model.
9	Fan modules	Four 80 mm (3.14") x 56 mm (2.20") variable-speed fans
10	PDB assembly	Two power distribution boards (2U4N) and single cage
11	Power distribution board assembly cover	Remove to service the PDB assembly

3.5" HDD System

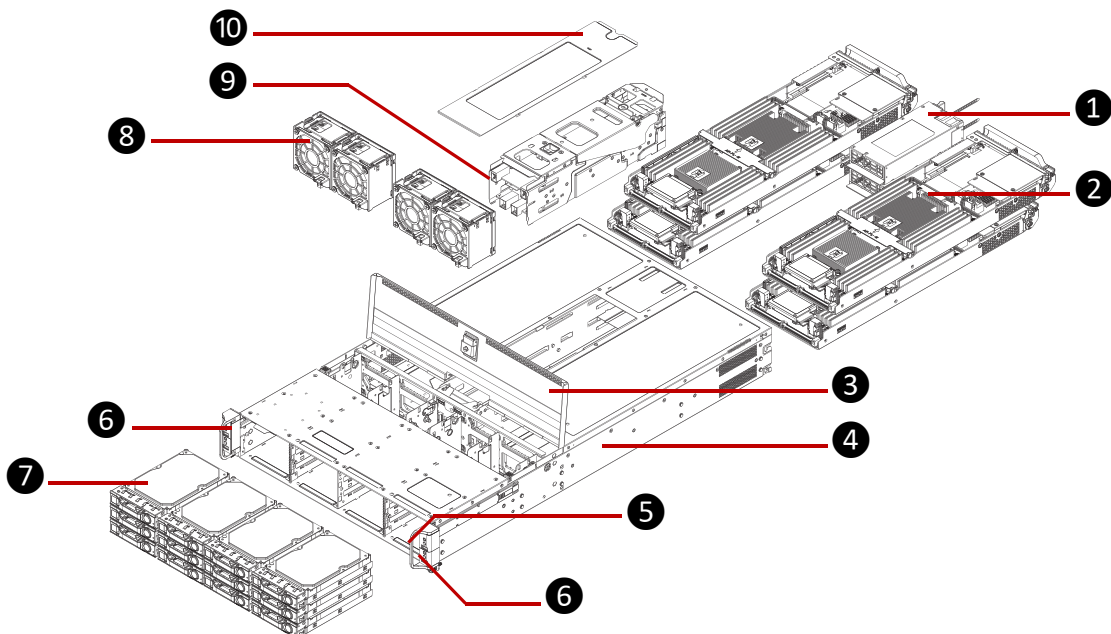


Figure 1-2. 3.5" System Component Overview

Table 3: 3.5" System Component Overview

No.	ITEM	DESCRIPTION
1	Power supply unit	Two 1600W high efficiency redundant PSU, 200-240Vac, 50/60Hz, 10A or 240Vdc, 10A
		Two 2200W high efficiency redundant PSU, 100-120Vac, 50/60Hz, 15.5A Max or 240Vdc, 14A Max
2	MB sled	Hot-swap mainboard sleds (x4)
3	Fan assembly cover	Open to service fan modules
4	Chassis	2U System chassis 444mm/17.48"(W)x 87.5mm/3.44"(H)x 790mm/31.1"(D)

Table 3: 3.5" System Component Overview (Continued)

No.	ITEM	DESCRIPTION
5	Server handle	Two server handles used for pulling the system out of the rack CAUTION! THE HANDLES ARE DESIGNED FOR THE EXTENSION OF THE SYSTEM FROM THE RACK. THE HANDLES ARE NOT DESIGNED TO CARRY THE WEIGHT OF THE SYSTEM. DO NOT USE THE HANDLES TO MOVE OR LIFT THE SYSTEM.
6	Front control panels	Front control panels 1/2 (left) and 3/4 (right). See <i>Front Control Panel and Status LED</i> on page 1-9 for further information
7	3.5" HDD bay	3.5" hard disk drive (HDD) modules (x12)
8	Fan modules	Four 80 mm (3.14") x 56 mm (2.20") variable-speed fans
9	PDB assembly	Two power distribution boards (2U4N) and single cage
10	Power distribution board cover	Remove to service the PDB assembly

System Front View

2.5" HDD SKU

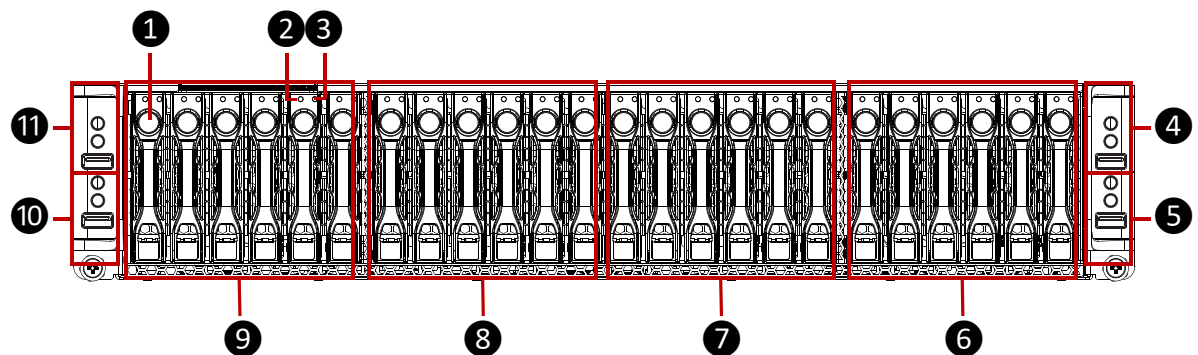


Figure 1-3. Front View for 2.5" HDD Configuration

Table 4: System Front View for 2.5" HDD Configuration

No	FEATURE	DESCRIPTION
1	HDD release lever & button	Button and release lever for removing HDD tray
2	HDD locate LED	Blue On: Healthy Amber On: Warning
3	HDD active LED	Blue On: Access
4	Front control panel (Node 3)	Includes power/ID button, power/ID/status LED, and USB port
5	Front control panel (Node 4)	

Table 4: System Front View for 2.5" HDD Configuration (Continued)

No	FEATURE	DESCRIPTION
6	Node 4	MB4 HDD Bay
7	Node 3	MB3 HDD Bay
8	Node 2	MB2 HDD Bay
9	Node 1	MB1 HDD Bay
10	Front control panel (Node 2)	Includes power/ID button, power/ID/status LED, and USB port
11	Front control panel (Node 1)	

3.5" HDD SKU

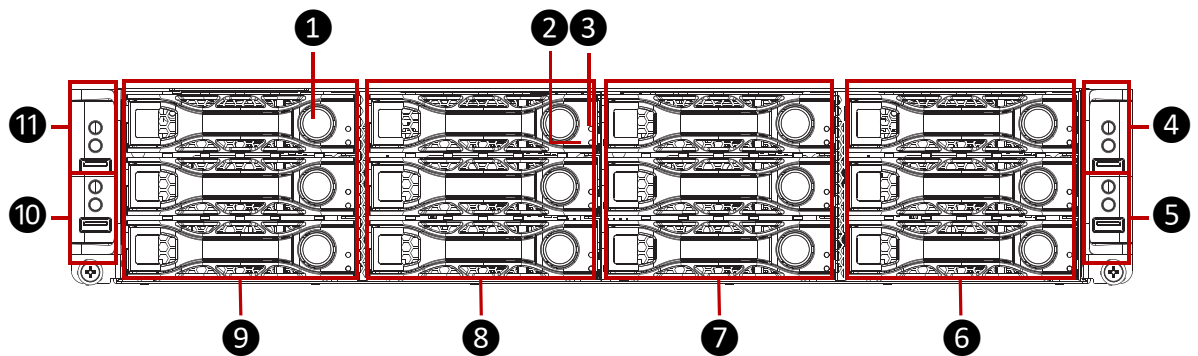


Figure 1-4. Front View for 3.5" HDD Configuration

Table 5: System Front View for 3.5" HDD Configuration

No.	FEATURE	DESCRIPTION
1	HDD release lever & button	Button and release lever for removing HDD tray
2	HDD active LED	Blue: Access
3	HDD locate LED	Blue: Healthy Amber: Warning
4	Front control panel (Node 3)	Includes power/ID button, power/ID/status LED, and USB port
5	Front control panel (Node 4)	
6	Node 4	MB4 HDD Bay
7	Node 3	MB3 HDD Bay
8	Node 2	MB2 HDD Bay
9	Node 1	MB1 HDD Bay

Table 5: System Front View for 3.5" HDD Configuration (Continued)

No.	FEATURE	DESCRIPTION
10	Front control panel (Node 2)	Includes power/ID button, power/ID/status LED, and USB port
11	Front control panel (Node 1)	

Front Control Panel and Status LED

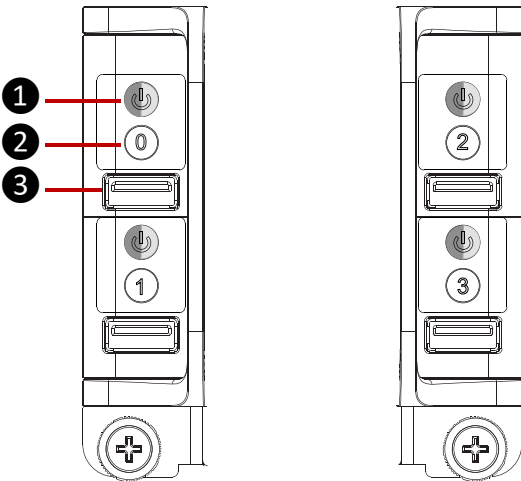


Figure 1-5. Front Control Panel

Table 6: Front Control Panel

No	FEATURE	DESCRIPTION
1	Power button with LED	Blue on: System power on Off: System power off Amber blinking: DC off and fault Amber and blue blinking: DC on and fault
2	ID button with LED	Blue blinking: ID indicator Off: No indicator 1: MB1 (2: MB2, 3: MB3, 4: MB4)
3	USB 2.0 port	USB 2.0 port

Rear View

The system rear consists of four mainboard sleds and two power supply units. Dummy modules are installed to maintain proper airflow in the event that mainboard slots are not fully populated.

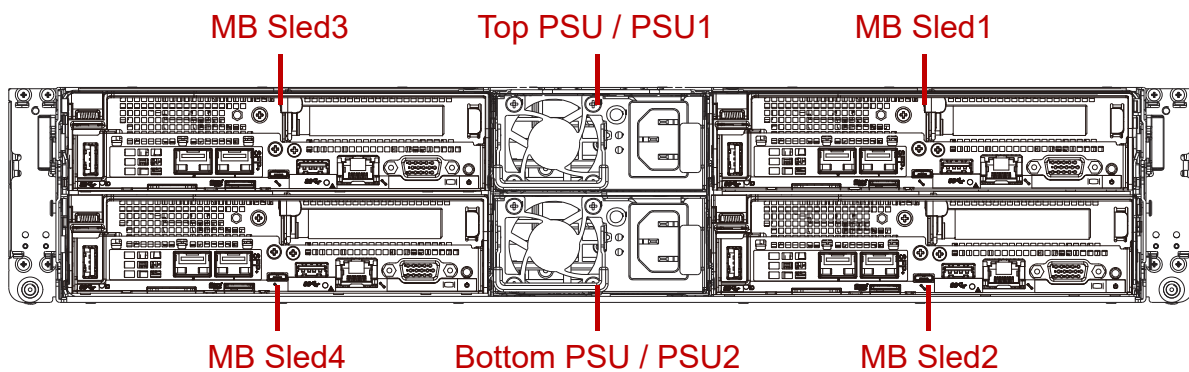


Figure 1-6. System Rear View

Rear I/O Ports

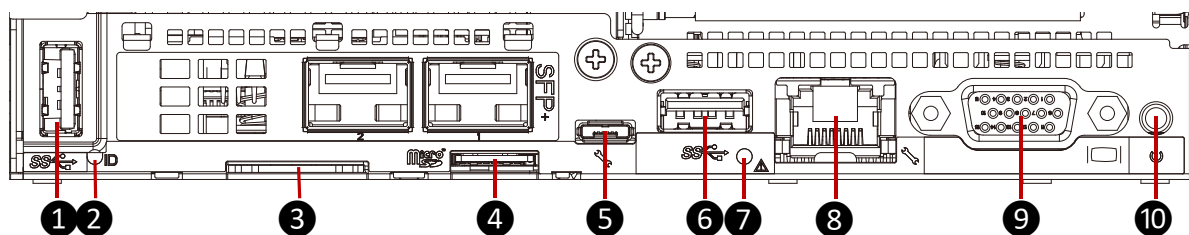


Figure 1-7. Rear I/O Ports

Table 7: Rear I/O Ports

No	FEATURE	DESCRIPTION
1	USB 3.0 port	Connect to USB device
2	ID LED	Light for identification Blue blinking: Identifier
3	Asset tag	Record serial number or other important information
4	MicroSD slot	Backup BMC System Event Log (SEL)
5	Micro-USB port	Connect to UART cable for RS232 connection
6	USB 3.0 port	Connect to USB device
7	Fault LED	Amber blinking: BMC log critical event Off: normal
8	Management port	Dedicated BMC LAN port (RJ45) for remote control / management.
9	VGA port	Standard VGA compatible, 15-pin connector supporting up to 1920 x 1200 32bpp@60Hz resolution. Connect to display device.

Table 7: Rear I/O Ports (Continued)

No	FEATURE	DESCRIPTION
10	Power button with LED	Press and hold more than four seconds to turn off the motherboard. Off: power off Blue blinking: BMC reports system health fault

Note:

The USB connector must be:

No bigger than 17mm/ 0.67" (W) x 8 mm / 0.31" (H) to avoid interference with the other ports.

Note:

The temperature through the ventilation holes was high. Make sure to use a cable rated to withstand 75 degrees or higher.

HDD Configuration

The multi-node systems are available as a 2.5" or 3.5" SKU. Within the HDD array, the HDD enumeration is as follows:

2.5" HDD Configuration

Note:

HDD slots 4/5 for MB1, HDD slots 10/11 for MB2, HDD slots 16/17 for MB3 and HDD slots 22/23 for MB4 are designed to support Tri-Mode (SATA/SAS/SFF8639=U.2) only for model T42SP-2U

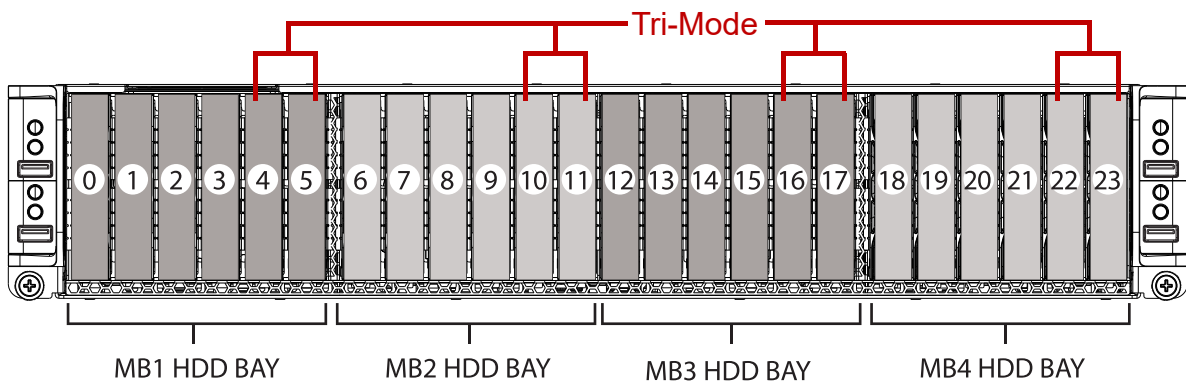


Figure 1-8. 2.5" HDD Configuration

Table 8: Intel® VMD PCIe Root Port BIOS Setup Option Table*

HDD SLOT#	MAINBOARD SLED #	MODEL: T42SP-2U
4	1	CPU Socket 0's VMD Port 2A
5		CPU Socket 0's VMD Port 2B

Table 8: Intel® VMD PCIe Root Port BIOS Setup Option Table* (Continued)

HDD SLOT#	MAINBOARD SLED #	MODEL: T42SP-2U
10	2	CPU Socket 0's VMD Port 2A
11		CPU Socket 0's VMD Port 2B
16	3	CPU Socket 0's VMD Port 2A
17		CPU Socket 0's VMD Port 2B
22	4	CPU Socket 0's VMD Port 2A
23		CPU Socket 0's VMD Port 2B

*See *Enabling the Intel® VMD technology* on page 1-15 for more information.

3.5" HDD Configuration

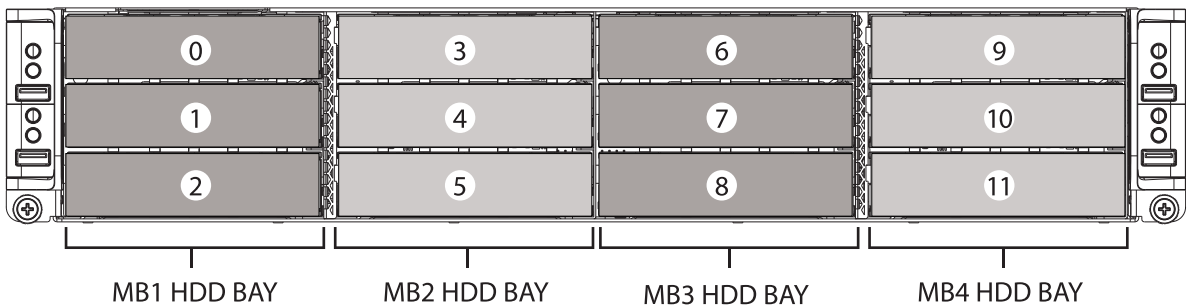


Figure 1-9. 3.5" HDD Configuration

LAN LEDs

Power Sub-System

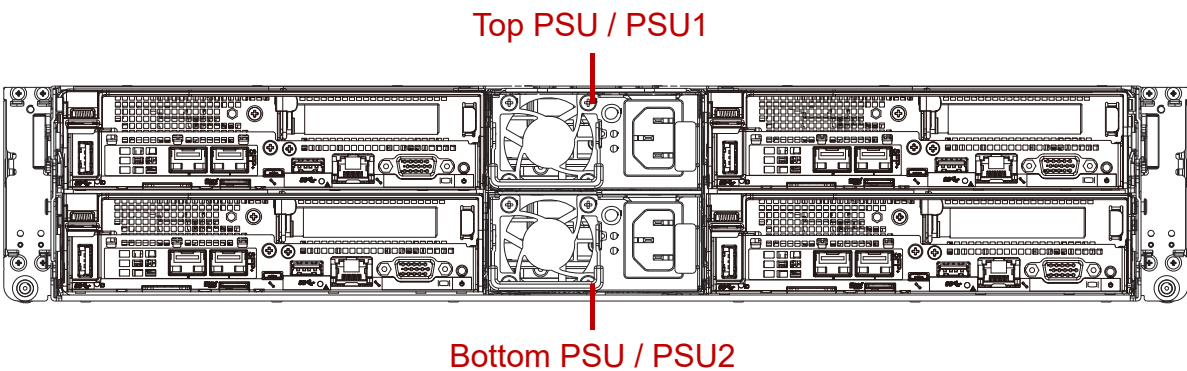


Figure 1-10. PSU to Mainboard Sled Description



WARNING!

Mainboard combinations other than the system are not supported.

A system has two modular Power Supply Units (PSU). Both PSUs are directly connected to the Power Distribution Boards (PDBs), the HDD backplane, and middle plane allowing each PSU to individually provide power for all mainboards (MB1 to MB4).

The power supply units supported: 2 x 1600W redundant PSU @ 200-240Vac, 50/60Hz, 10A or 240Vdc, 10A or 2 x 2200W redundant PSU @ 100-120Vac, 50/60Hz, 15.5A Max or 240Vdc, 14A Max.

PSU LED

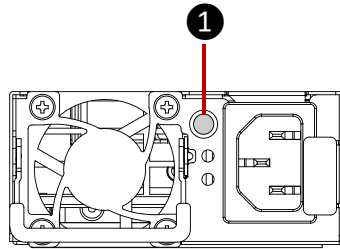


Figure 1-11. PSU LED

Table 9: PSU LED Description

STATUS	DESCRIPTION
Green	Normal operation
Amber: blinking	Main power fault and stop DC output
Green: blinking@0.5Hz	AC present and only standby power on
Green: blinking@2Hz	Sleep in smart redundant and offline mode

Cooling Sub-System

Fans may spin for some time after the system has been powered off. Allow time for the fans to stop rotating before handling system components.

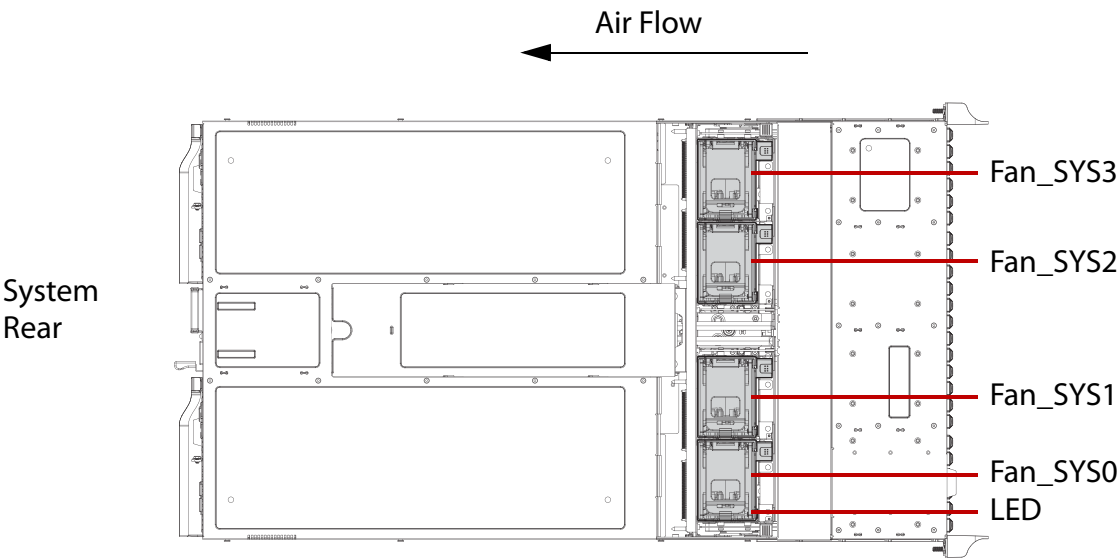


Figure 1-12. Cooling Sub System (2.5" HDD model shown)

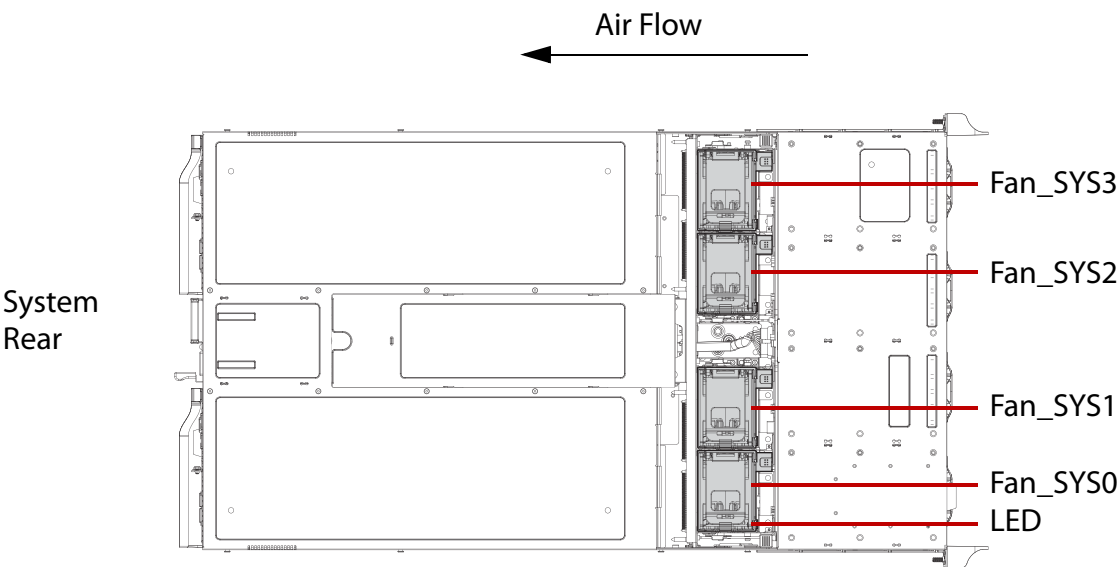


Figure 1-13. Cooling Sub System (3.5" HDD model shown)

Table 10: Fan LED Description

STATUS	DESCRIPTION
Red	Fan removed and stop
Off	Fan present

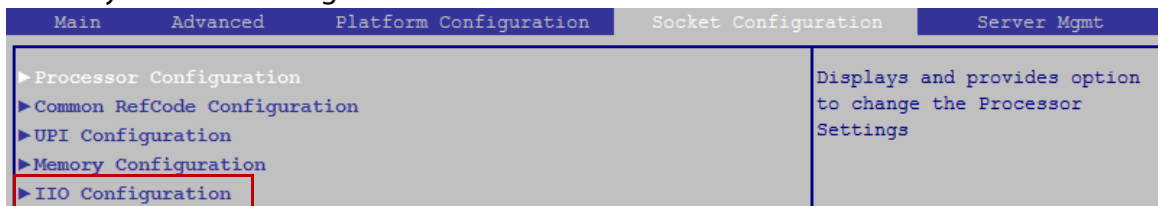
Enabling the Intel® VMD technology

Note:

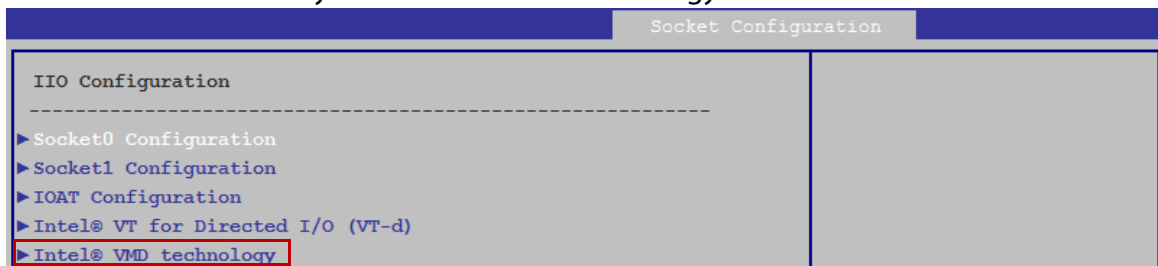
You can use Intel® VROC HII to create the RAID volume. These volume can be created at any point before or after your system is successfully running your Windows installation, but may not be used as the boot volume.

To enable the Intel VMD technology feature, you might need to select “Intel® VMD technology” in “IIO configuration” under the “Socket Configuration” screen from BIOS Main screen by pressing the right arrow until the Socket Configuration screen is chosen.

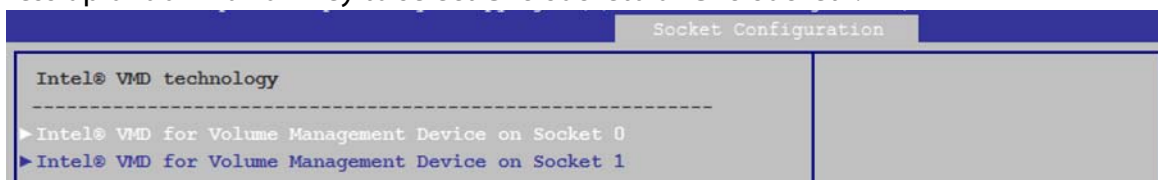
Press the right arrow key until “Socket Configuration” is selected and press the down arrow key until “IIO Configuration” is selected.



Press the down arrow key until “Intel® VMD technology” is selected.



Press up or down arrow key to select CPU socket0 or CPU socket1.



Then you can enable/disable the Intel® VMD feature for your system.

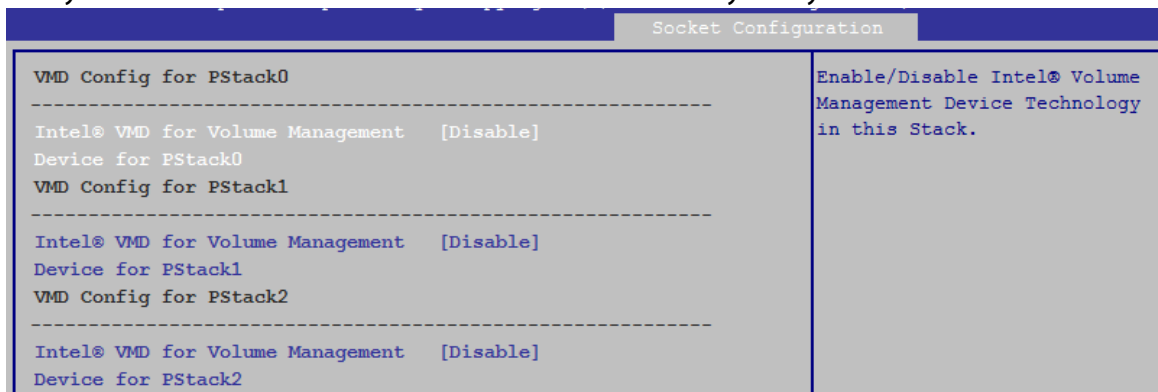


Table 11: Intel® VMD on Socket X Screen Description

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
VMD Config for PStack0			Information only.
Intel® VMD for Volume Management Device for PStack0	[Disable] [Enable]	Enable/Disable Intel® Volume Management Device Technology in this Stack.	
VMD Config for PStack1			
Intel® VMD for Volume Management Device for PStack1	[Disable] [Enable]	Enable/Disable Intel® Volume Management Device Technology in this Stack.	
VMD Config for PStack2			
Intel® VMD for Volume Management Device for PStack2	[Disable] [Enable]	Enable/Disable Intel® Volume Management Device Technology in this Stack.	

Enabling Platform Trusted Technology (PTT)

To enable platform trusted technology feature, you might need to select “Server ME Configuration” in “Platform Configuration” screen from BIOS Main screen by pressing the right arrow until the Platform Configuration screen is chosen. Press the down arrow key until “Server ME Configuration” is selected.

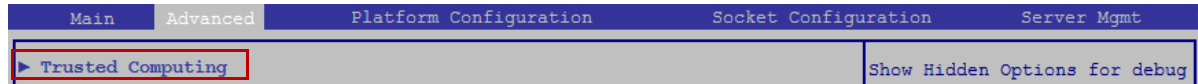
Main	Advanced	Platform Configuration	Socket Configuration	Server Mgmt
▶ PCH Configuration ▶ Miscellaneous Configuration ▶ Server ME Configuration			Displays and provides option to change the PCH Settings	

Press down arrow key to select PTT Support. Then you can enable/disable the PTT feature for your system

Platform Configuration		
General ME Configuration		Enable/disable Platform Trusted Technology (PTT) support
Oper. Firmware Version	0A:4.0.3.113	
Backup Firmware Version	N/A	
Recovery Firmware Version	0A:4.0.3.113	
ME Vendor Label	0x0000007	
ME Firmware Status #1	0x00F0382	
ME Firmware Status #2	0x36520006	
Current State	Operation	
Error Code	No Error	
Recovery Cause	N/A	
PTT Support	[Enable]	

TPM configuration settings

To setup the TPM configuration, you might need to select “Trusted Computing” in “Advanced” screen from BIOS Main screen by pressing the right arrow until the Advanced screen is chosen. Press the down arrow key until “Trusted Computing” is selected.



Now you can setup the TPM configuration.

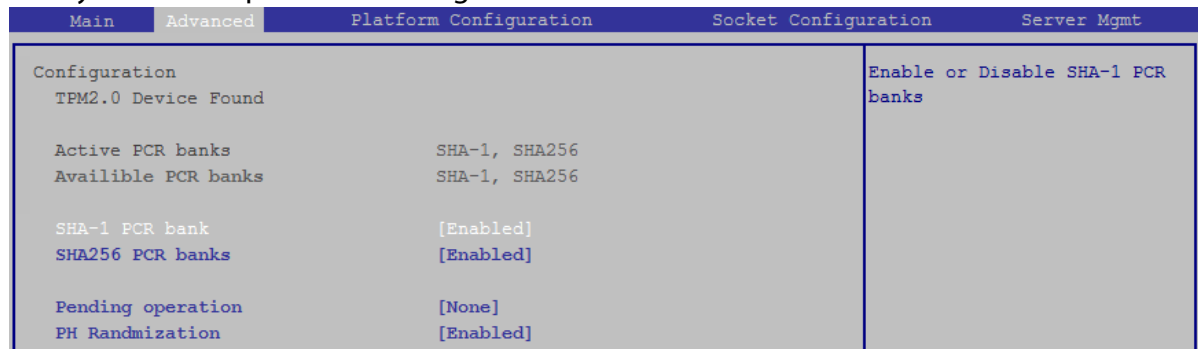


Table 12: Configuration on Trusted Computing Screen Description

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
TPM 2.0 Device Found			Information only. Displays current TPM Status.
Active PCR banks			Information only. Displays current active PCR banks.
Available PCR banks			Information only. Displays current available PCR banks.
SHA-1 PCR Bank	[Disabled] [Enabled]	Enable or Disable SHA-1 PCR Bank	Only appears when “Available PCR banks” includes this type.
SHA256 PCR Bank	[Disabled] [Enabled]	Enable or Disable SHA256 PCR Bank	Only appears when “Available PCR banks” includes this type.
Pending operation	[None] [TPM Clear]	Enable/Disable Security Device. NOTE: Your computer will reboot during restart in order to change state of the device.	Only appears when TPM device exist
PH Randomization	[Disabled] [Enabled]	Enables or Disables Platform Hierachy randomization.	Only appears when TPM device exist

Regulatory and Compliance Information

Chapter 2

This section provides regulatory and compliance information applicable to this system.

Server Safety Information

To reduce the risk of bodily injury, electrical shock, fire, and equipment damage, read this document and observe all warnings and precautions in this guide before installing or maintaining your server product.

In the event of a conflict between the information in this document and information provided with the product or on the website for a particular product, the product documentation takes precedence.

Your server should be integrated and serviced only by technically qualified persons.

You must adhere to the guidelines in this guide and the assembly instructions in your server manuals to ensure and maintain compliance with existing product certifications and approvals. Use only the described, regulated components specified in this guide. Use of other products / components will void the UL Listing and other regulatory approvals of the product, and may result in noncompliance with product regulations in the region(s) in which the product is sold.

Safety Warnings and Cautions

To avoid personal injury or property damage, before you begin installing the product, read, observe, and adhere to all of the following safety instructions and information. The following safety symbols may be used throughout the documentation and may be marked on the product and / or the product packaging.

CAUTION	Indicates the presence of a hazard that may cause minor personal injury or property damage if the CAUTION is ignored.
WARNING	Indicates the presence of a hazard that may result in serious personal injury if the WARNING is ignored.
	Indicates potential hazard if indicated information is ignored.
	Indicates shock hazards that result in serious injury or death if safety instructions are not followed.
	Indicates hot components or surfaces.
	Indicates do not touch fan blades, may result in injury.
	Indicates to unplug all AC power cord(s) to disconnect AC power.
	Please recycle battery.
	The rail racks are designed to carry only the weight of the server system. Do not use rail-mounted equipment as a workspace. Do not place additional load onto any rail-mounted equipment.
	Indicates four or more people are required to safely handle the system.

Intended Application Uses

This product was evaluated as Information Technology Equipment (ITE), which may be installed in offices, schools, computer rooms, and similar commercial type locations. The suitability of this product for other product categories and environments (such as medical, industrial, residential, alarm systems, and test equipment), other than an ITE application, may require further evaluation.

Site Selection

The system is designed to operate in a typical office environment. Choose a site that is:

- Clean, dry, and free of airborne particles (other than normal room dust).
- Well-ventilated and away from sources of heat including direct sunlight and radiators.
- Away from sources of vibration or physical shock.
- Isolated from strong electromagnetic fields produced by electrical devices.
- In regions that are susceptible to electrical storms, we recommend you plug your system into a surge suppressor and disconnect telecommunication lines to your modem during an electrical storm.
- Provided with a properly grounded wall outlet.
- Provided with sufficient space to access the power supply cord(s), because they serve as the product's main power disconnect.
- Provided with either two independent AC power sources or two independent phases from a single source.

Equipment Handling Practices

Reduce the risk of personal injury or equipment damage:

- Conform to local occupational health and safety requirements when moving and lifting equipment.
- Use mechanical assistance or other suitable assistance when moving and lifting equipment.
- To reduce the weight for easier handling, remove any easily detachable components.

Power and Electrical Warnings

Caution: The power button, indicated by the stand-by power marking, DOES NOT completely turn off the system AC power, 5V standby power is active whenever the system is plugged in. To remove power from system, you must unplug the AC power cord from the wall outlet. Your system may use more than one AC power cord. Make sure all AC power cords are unplugged. Make sure the AC power cord(s) is / are unplugged before you open the chassis, or add or remove any non hot-plug components.

Do not attempt to modify or use an AC power cord if it is not the exact type required. A separate AC cord is required for each system power supply.

Some power supplies in servers use Neutral Pole Fusing. To avoid risk of shock use caution when working with power supplies that use Neutral Pole Fusing.

The power supply in this product contains no user-serviceable parts. Do not open the power supply. Hazardous voltage, current and energy levels are present inside the power supply. Return to manufacturer for servicing.

When replacing a hot-plug power supply, unplug the power cord to the power supply being replaced before removing it from the server.

To avoid risk of electric shock, turn off the server and disconnect the power cord, telecommunications systems, networks, and modems attached to the server before opening it.

Power Cord Warnings

If an AC power cord was not provided with your product, purchase one that is approved for use in your country.

Only use certified AC power cord to connect to both the power distribution unit (PDU) and server system installed in your rack.

Caution: *To avoid electrical shock or fire, check the power cord(s) that will be used with the product as follows:*

- *Do not attempt to modify or use the AC power cord(s) if they are not the exact type required to fit into the grounded electrical outlets.*
- *The power cord(s) must meet the following criteria:*
 - *The power cord must have an electrical rating that is greater than that of the electrical current rating marked on the product.*
 - *The power cord must have safety ground pin or contact that is suitable for the electrical outlet.*
 - *The power supply cord(s) is / are the main disconnect device to AC power. The socket outlet(s) must be near the equipment and readily accessible for disconnection.*
 - *The power supply cord(s) must be plugged into socket-outlet(s) that is /are provided with a suitable earth ground.*

System Access Warnings

Caution: *To avoid personal injury or property damage, the following safety instructions apply whenever accessing the inside of the product:*

- *Turn off all peripheral devices connected to this product.*
- *Turn off the system by pressing the power button to off.*
- *Disconnect the AC power by unplugging all AC power cords from the system or wall outlet.*
- *Disconnect all cables and telecommunication lines that are connected to the system.*
- *Retain all screws or other fasteners when removing access cover(s). Upon completion of accessing inside the product, refasten access cover with original screws or fasteners.*
- *Do not access the inside of the power supply. There are no serviceable parts in the power supply. Return to manufacturer for servicing.*
- *Power down the server and disconnect all power cords before adding or replacing any non hot-plug component.*
- *When replacing a hot-plug power supply, unplug the power cord to the power supply being replaced before removing the power supply from the server.*

Caution: *If the server has been running, any installed processor(s) and heat sink(s) may be hot.*

Unless you are adding or removing a hot-plug component, allow the system to cool before opening the covers. To avoid the possibility of coming into contact with hot component(s) during a hot-plug installation, be careful when removing or installing the hot-plug component(s).

Caution: *To avoid injury do not contact moving fan blades. If your system is supplied with a guard over the fan, do not operate the system without the fan guard in place.*

Rack Mount Warnings

Note: *The following installation guidelines are required by UL for maintaining safety compliance when installing your system into a rack.*

The equipment rack must be anchored to an unmovable support to prevent it from tipping when a server or piece of equipment is extended from it. The equipment rack must be installed according to the rack manufacturer's instructions.

Install equipment in the rack from the bottom up, with the heaviest equipment at the bottom of the rack.

Extend only one piece of equipment from the rack at a time.

You are responsible for installing a main power disconnect for the entire rack unit. This main disconnect must be readily accessible, and it must be labeled as controlling power to the entire unit, not just to the server(s).

To avoid risk of potential electric shock, a proper safety ground must be implemented for the rack and each piece of equipment installed in it.

Elevated Operating Ambient - If installed in a closed or multi-unit rack assembly, the operating ambient temperature of the rack environment may be greater than room ambient. Therefore, consideration should be given to installing the equipment in an environment compatible with the maximum ambient temperature (T_{ma}) specified by the manufacturer.

Reduced Air Flow - Installation of the equipment in a rack should be such that the amount of air flow required for safe operation of the equipment is not compromised.

Mechanical Loading - Mounting of the equipment in the rack should be such that a hazardous condition is not achieved due to uneven mechanical loading.

Circuit Overloading - Consideration should be given to the connection of the equipment to the supply circuit and the effect that overloading of the circuits might have on overcurrent protection and supply wiring. Appropriate consideration of equipment nameplate ratings should be used when addressing this concern.

Reliable Earthing - Reliable earthing of rack-mounted equipment should be maintained.

Particular attention should be given to supply connections other than direct connections to the branch circuit (e.g. use of power strips).

Electrostatic Discharge (ESD)

Caution: *ESD can damage drives, boards, and other parts. We recommend that you perform all procedures at an ESD workstation. If one is not available, provide some ESD protection by wearing an antistatic wrist strap attached to chassis ground -- any unpainted metal surface -- on your server when handling parts.*

Always handle boards carefully. They can be extremely sensitive to ESD. Hold boards only by their edges. After removing a board from its protective wrapper or from the server, place the board component side up on a grounded, static free surface. Use a conductive foam pad if available but not the board wrapper. Do not slide board over any surface.

Other Hazards

Battery Replacement

Caution: *There is the danger of explosion if the battery is incorrectly replaced. When replacing the battery, use only the battery recommended by the equipment manufacturer.*

Dispose of batteries according to local ordinances and regulations.

Do not attempt to recharge a battery.

Do not attempt to disassemble, puncture, or otherwise damage a battery.

Cooling and Airflow

Caution: Carefully route cables as directed to minimize airflow blockage and cooling problems. For proper cooling and airflow, operate the system only with the chassis covers installed. Operating the system without the covers in place can damage system parts. To install the covers:

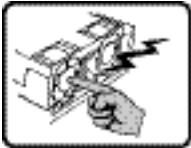
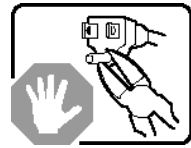
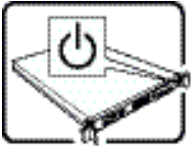
- Check first to make sure you have not left loose tools or parts inside the system.
- Check that cables, add-in cards, and other components are properly installed.
- Attach the covers to the chassis according to the product instructions.

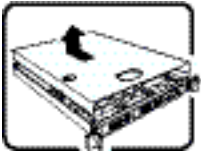
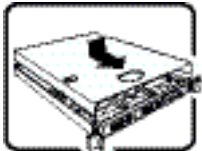
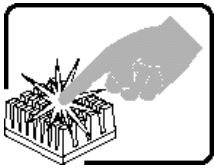
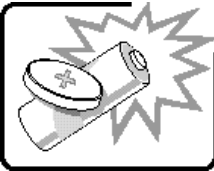
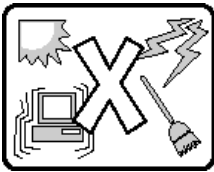
Laser Peripherals or Devices

Caution: To avoid risk of radiation exposure and / or personal injury:

- Do not open the enclosure of any laser peripheral or device
- Laser peripherals or devices have are not serviceable
- Return to manufacturer for servicing

Use certified Optical Fiber Transceiver Class I Laser Product

	The power supply in this product contains no user-serviceable parts. Refer servicing only to qualified personnel.
	Do not attempt to modify or use the supplied AC power cord if it is not the exact type required. A product with more than one power supply will have a separate AC power cord for each supply.
	The power button on the system does not turn off system AC power. To remove AC power from the system, you must unplug each AC power cord from the wall outlet or power supply. The power cord(s) is considered the disconnect device to the main (AC) power. The socket outlet that the system plugs into shall be installed near the equipment and shall be easily accessible.
	SAFETY STEPS: Whenever you remove the chassis covers to access the inside of the system, follow these steps: 1. Turn off all peripheral devices connected to the system. 2. Turn off the system by pressing the power button. 3. Unplug all AC power cords from the system or from wall outlets. 4. Label and disconnect all cables connected to I/O connectors or ports on the back of the system. 5. Provide some electrostatic discharge (ESD) protection by wearing an antistatic wrist strap attached to chassis ground of the system-any unpainted metal surface-when handling components. 6. Do not operate the system with the chassis covers removed.

	After you have completed the six SAFETY steps above, you can remove the system covers. To do this: 1. Unlock and remove the padlock from the back of the system if a padlock has been installed. 2. Remove and save all screws from the covers. 3. Remove the cover(s).
	For proper cooling and airflow, always reinstall the chassis covers before turning on the system. Operating the system without the covers in place can damage system parts. To install the covers: 1. Check first to make sure you have not left loose tools or parts inside the system. 2. Check that cables, add-in cards, and other components are properly installed. 3. Attach the covers to the chassis with the screws removed earlier, and tighten them firmly. 4. Insert and lock the padlock to the system to prevent unauthorized access inside the system. 5. Connect all external cables and the AC power cord(s) to the system.
	A microprocessor and heat sink may be hot if the system has been running. Also, there may be sharp pins and edges on some board and chassis parts. Contact should be made with care. Consider wearing protective gloves.
	Danger of explosion if the battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the equipment manufacturer. Dispose of used batteries according to manufacturer's instructions.
	The system is designed to operate in a typical office environment. Choose a site that is: • Clean and free of airborne particles (other than normal room dust). • Well ventilated and away from sources of heat including direct sunlight. • Away from sources of vibration or physical shock. • Isolated from strong electromagnetic fields produced by electrical devices. • In regions that are susceptible to electrical storms, we recommend you plug your system into a surge suppressor and disconnect telecommunication lines to your modem during an electrical storm. • Provided with a properly grounded wall outlet. • Provided with sufficient space to access the power supply cord(s), because they serve as the product's main power disconnect.

	The rail racks are designed to carry only the weight of the server system. Do not use rail-mounted equipment as a workspace. Do not place additional load onto any rail-mounted equipment.
	Heavy object. Indicates four or more people are required to safely handle the system.

Product Regulatory Compliance Markings

This product is marked with the following Product Certification Markings:

Product Regulatory Compliance Markings

Regulatory Compliance	Region	Marking
cULus Listing Mark	USA / Canada	
CE Mark	Europe	
FCC Marking (Class A)	USA	This device complies with Part 15 of the FCC Rules. Operation of this device is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) This device must accept any interference received, including interference that may cause undesired operation.
ICES	Canada	CAN ICES-3 (A)/NMB-3(A)
VCCI Marking (Class A)	Japan	この装置は、クラス A 情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。 VCCI-A
BSMI Certification Number & Class A Warning	Taiwan	 警告使用者： 此為甲類資訊技術設備，於居住環境使用中時，可能會造成射頻擾動，在此種情況下，使用者會被要求採取某些適當的對策。
Recycling Package Mark	Other than China	 
MSIP	Korea	 A급 기기 (업무용 정보통신기기) 이 기기는 업무용으로 전자파적합등록을 한 기기이오니 판매자 또는 사용자는 이 점을 주의하시기 바라며, 만약 잘못판매 또는 구입하였을 때에는 가정용으로 교환하시기 바랍니다.
RCM	Australia	

Electromagnetic Compatibility Notices

FCC Verification Statement (USA)

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Europe (CE Declaration of Conformity)

This product has been tested in accordance too, and complies with the Low voltage Directive (2006/95/EC) and EMC Directive (2004/108/EC). The product has been marked with the CE Mark to illustrate its compliance.

VCCI (Japan)

この装置は、クラス A 情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。

VCCI-A

English translation of the notice above:

This is a Class A product based on the standard of the Voluntary Control Council for Interference (VCCI) from Information Technology Equipment. If this is used near a radio or television receiver in a domestic environment, it may cause radio interference. Install and use the equipment according to the instruction manual.

BSMI (Taiwan)

The BSMI Certification Marking and EMC warning is located on the outside rear area of the product.

警告使用者：

此為甲類資訊技術設備，於居住環境中使用時，可能會造成射頻擾動，在此種情況下，使用者會被要求採取某些適當的對策。

MSIP (Korea)

Ministry of Science, ICT & Future Planning (MSIP) Class A Statement:

A 급 기기(업무용 정보통신기기)

이 기기는 업무용으로 전자파적합등록을 한 기기이오니 판매자 또는 사용자는 이 점을 주의하시기 바라며, 만약 잘못 판매 또는 구입하였을 때에는 가정용으로 교환하시기 바랍니다.

English translation of the notice above:

“A” class equipment (info / technology equipment for business purpose)

As this equipment has undergone EMC registration for business purpose, the seller and/or the buyer is asked to beware of this point and in case a wrongful sale or purchase has been made, it is asked that a change to household use be made.

Regulated Specified Components

To maintain the UL listing and compliance to other regulatory certifications and/or declarations, the following regulated components must be used and conditions adhered to. Interchanging or use of other component will void the UL listing and other product certifications and approvals.

Updated product information for configurations can be found on the site at the following URL:

www.QCT.io

If you do not have access to the Web address, please contact your local representative.

- **Add-in cards:** must have a printed wiring board flammability rating of minimum UL94V-1. Add-in cards containing external power connectors and/or lithium batteries must be UL recognized or UL listed. Any add-in card containing modem telecommunication circuitry must be UL listed. In addition, the modem must have the appropriate telecommunications, safety, and EMC approvals for the region in which it is sold.
- **Peripheral Storage Devices:** must be UL recognized or UL listed accessory and TUV or VDE licensed. Maximum power rating of any one device is 19 watts. Total server configuration is not to exceed the maximum loading conditions of the power supply.

Restriction of Hazardous Substances (RoHS) Compliance

Quanta® Computer Inc. has a system in place to restrict the use of banned substances in accordance with the European Directive 2011/65/EU. Compliance is based on declaration that materials banned in the RoHS Directive are either (1) below all applicable threshold limits or (2) an approved / pending RoHS exemption applies.

RoHS implementation details are not fully defined and may change.

Threshold limits and banned substances are noted below:

- Quantity limit of 0.1% by mass (1000 PPM) for:
 - Lead
 - Mercury
 - Hexavalent Chromium
 - Polybrominated Diphenyl Ethers (PBDE)
- Quantity limit of 0.01% by mass (100 PPM) for:
 - Cadmium

End of Life / Product Recycling

Product recycling and end-of-life take-back systems and requirements vary by country. Contact the retailer or distributor of this product for information about product recycling and / or take-back.

設備名稱： 伺服器 ，型號（型式）： T42S-2U/T42SP-2U
Equipment name Type designation (Type)

單元 Unit	限用物質及其化學符號 Restricted substances and its chemical symbols					
	鉛Lead (Pb)	汞Mercury (Hg)	鎘Cadmium (Cd)	六價鉻 Hexavalent chromium (Cr ⁺⁶)	多溴聯苯 Polybrominated biphenyls (PBB)	多溴二苯醚 Polybrominated diphenyl ethers (PBDE)
金屬機殼	○	○	○	○	○	○
電路板	○	○	○	○	○	○
塑膠面板	○	○	○	○	○	○
電源線、 連接線	○	○	○	○	○	○

備考1. “超出0.1 wt %” 及 “超出0.01 wt %” 係指限用物質之百分比含量超出百分比含量基準值。

Note 1 : “Exceeding 0.1 wt %” and “exceeding 0.01 wt %” indicate that the percentage content of the restricted substance exceeds the reference percentage value of presence condition.

備考2. “○” 係指該項限用物質之百分比含量未超出百分比含量基準值。

Note 2 : “○” indicates that the percentage content of the restricted substance does not exceed the percentage of reference value of presence.

備考3. “—” 係指該項限用物質為排除項目。

Note 3 : The “—” indicates that the restricted substance corresponds to the exemption.